

Location: `Swetha\_StrongHER\_projects/6G\_AI\_Channel\_Estimation/code`

This repository is a runnable, 3GPP-aligned AI-native channel intelligence platform. It generates a telecom-grade synthetic dataset (60,000 rows), trains ten cooperating agents with explicit train / validation / test sp

Traditional LS/MMSE estimators degrade in THz, RIS, ultra-massive MIMO, NTN, and high-Doppler 6G channels, inflate pilot overhead, cannot predict CSI, and are exposed to pilot contamination, jamming, spoofing, poisoning,

1. 3GPP CDL/TDL channel modeling (TR 38.901) plus NTN (TR 38.811) and THz/RIS research ranges

3. Architecture

Seven layers from the research document are implemented as code modules:

|-----|-----|

4. Dataset (synthetic, 3GPP-aligned)

Radio: fc, SCS, bandwidth, Nt/Nr, delay spread, Doppler, SNR/SINR, RSRP/RSRQ/RSSI, CQI, AoA/AoD

Attacks: normal, pilot contamination, jamming, CSI spoofing, false CSI injection, poisoning, adversarial, backdoor

Scenario counts: `{'UMi': 8214, 'THz': 8195, 'FR2-mmWave': 8129, 'RMa': 8104, 'UMa': 8100, 'RIS': 8092, 'NTN-LEO': 8089, 'InH': 8077}`

Related tables: `mobility\_dataset.csv`, `security\_dataset.csv`, `digital\_twin\_states.csv`.

Every ML agent is fit on train only. Hyperparameter-free scores are reported on validation and test (15% each, seed 42).

|-----|-----|-----|

| csi_prediction | yes | test_r2=0.9969, test_rmse=0.009642, test_mean_accuracy=0.8905 |

| mitigation | yes | test_accuracy=0.7843, test_success_rate=0.9607 |

| mobility | yes | test_accuracy=0.9652, test_ho_success=0.9983 |

| digital_twin | yes | test_r2=0.9975, test_mean_fidelity=0.9309 |

| orchestrator | yes | test_mean_nmse_ai=0.056116, test_mean_nmse_mmse=0.090862 |

Plots: `outputs/plots/model\_train\_val\_test.png`, `model\_training\_curves.png`, per-agent `outputs/plots/agents/\*\_hist\_cdf.png`.

Received signal model: $y = Hx + n$.

- MMSE: Wiener shrinkage using delay-spread correlation prior (TS 38.101-4 spirit)

Evaluation metrics match the research plan: MSE/RMSE/NMSE, BER, spectral efficiency, CSI prediction accuracy, attack precision/recall/F1/ROC-AUC.

`src/digital\_twin/channel\_twin.py` maintains cells, UEs, RIS panels, and LEO nodes. Each closed-loop step applies the orchestrator policy (hold / reduce pilots / increase pilots / mitigate) and updates NMSE, load, attack

python scripts/run_api_server.py

Tabs cover KPI overview, agent train/val/test, security classification, digital twin, plot gallery, knowledge sources, and chatbot. The chatbot answers NMSE, attacks, 3GPP/Nokia sources, and can run agents against the tw

- TR 38.811 NTN channel

- CFAM RP003187-2115 / RP003187-2929: DMRS-based PUSCH channel estimation (5GMax / 5G_L1_2794)

- RAN4#120 6G AI topic summary (testing)

Configured spaces were queried (`channel estimation`, DMRS, CSI, NRAC, RFSW). This client returned no indexed hits; Confluence status is recorded in `data/knowledge\_base/confluence\_references.json`.


```
`git clone https://scm.cci.nokia.net/AN/AI/agent-developer-artifacts/agent-shim.git` failed (HTTP Basic access denied). A local shim adapter lives in  
`src/shim/agent_shim_adapter.py`.
```

Dashboard styling follows Nokia CCFK FreeForm patterns used in the Autonomous RAN StrongHER project (`ccfk-dashboard/`` sources). The live demo also serves a self-contained operator dashboard that does not require the Nok

Descriptors: `src/oran/`.

```
cd Swetha_StrongHER_projects/6G_AI_Channel_Estimation/code
```


python scripts/run_end_to_end.py

Individual steps: `generate\_datasets.py`, `train\_validate\_test.py`, `generate\_visualizations.py`, `generate\_docs.py`, `generate\_slides.py`.

13. References

2. 3GPP TS 38.211, *NR; Physical channels and modulation*.

4. 3GPP TS 38.101-4, *UE radio transmission and reception; Part 4: Performance requirements*.

6. 3GPP TR 38.811, *Study on NR to support non-terrestrial networks*.

12. Project source documents: `6G\_AI\_CE\_01.docx`, `Swetha\_6G\_AI\_CE\_Proj\_01.pptx`.